AOU
الجامعة العربية المفتوحة
Arab Open University

## Faculty of Business Administration

# FIN301-F
# Financial & Securities Markets
## Version C

## Mid-term Examination
## Fall 2025-2026

## 3-December-2025

| | |
|---|---|
| **Number of Exam Pages: 3** (including this cover sheet) | **Time Allowed: 1.5 hours** |

## Instructions:

- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
- This is a **CLOSED** book examination. NO documents of any kind are allowed.
- You **MUST** answer the questions using only the answer booklets provided
- You **MUST** return the examination paper together with your answer booklets.
- Do **NOT** write your answers in pencil except graphs/sketches
- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatio
- Students are allowed to use ***non-programmable*** calculators only (if needed).
- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

AOU EXAM



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**Solve All questions**

***Question 1: (10 marks)***

Why is a share of *Apple* common stock an asset for its owner and a liability for Apple? Also, predict what will happen to interest rates if the public suddenly expects a large increase in stock prices.

***Question 2: (15 marks)***

Peter owns a $1,000-par zero-coupon bond that has 5 years of remaining maturity. He plans on selling the bond in one year and believes that the required yield next year will have the following probability distribution:

| Probability | Required Yield |
|---|---|
| 0.3 | 7.60% |
| 0.1 | 7.75% |
| 0.2 | 8.00% |
| 0.3 | 8.20% |
| 0.1 | 8.45% |

a. Calculate the expected price when you sell the bond

b. Calculate the standard deviation

***Question 3: (10 marks)***

1-year T-bill rates are expected to steadily increase by 400 basis points per year over the next 6 years. Determine the required interest rate on a 3-year T-bond and a 6-year T-bond if the current 1-year interest rate is 20%. Assume that the Pure Expectations Hypothesis for interest rates holds.

***Question 4: (15 marks)***

Suppose you want to earn an annualized discount rate of 4.5%. What would be the most you would pay for a 182-day Treasury bill that pays $40,000 at maturity?

**END OF QUESTION**

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# Formulas

Coupon Payment=

$$P = \frac{C}{(1+i)} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \dots + \frac{C}{(1+i)^n} + \frac{F}{(1+i)^n}$$

Expectation theory for n period bond=

$$i_{nt} = \frac{i_t + i_{t+1} + i_{t+2} + \dots + i_{t+(n-1)}}{n}$$

Treasury Bills discount rate=

$$i_{discount} = \frac{F - P}{F} \times \frac{360}{n}$$

Treasury bill investment rate=

$$i_{yt} = \frac{F - P}{P} \times \frac{365}{n}$$

Expected return= Sum(Prob × Price)

Variance= **Sum (Prob × (Price – Exp. Price)$^2$)**

Standard Deviation=$\sqrt{Prob(price - exp.price)^2}$

Liquidity Premium theory for n period bond=

$$i_{nt} = \frac{i_t + i^e_{t+1} + i^e_{t+2} + \dots + i^e_{t+(n-1)}}{n} + l_{nt}$$

Bond Rate of Return=

$$\frac{C + P_{t+1} - P_t}{P_t}$$

Pure Expectation theory for n period bond=

$$i_{nt} = \frac{i_t + i_{t+1} + i_{t+2} + \dots + i_{t+(n-1)}}{n}$$

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